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YEAR 12 · HSC COURSE

When petrol prices rise, do petrol thefts follow?

A research summary for **HSC Economics** — and a live case study in incentives, inflation and the difference between correlation and causation.

Adams, N. (2026). *How do petrol thefts respond to changing prices?* Crime and Justice Bulletin No. 280. Sydney: NSW Bureau of Crime Statistics and Research.

Between 2022 and 2024 the number of petrol thefts in NSW climbed sharply — but petrol prices over that period stayed relatively stable. In 2025 prices held steady again and thefts drifted down. That puzzle is the starting point for this study. If the old story ("prices up, thefts up") still held, the two lines should have moved together. They didn't. So the researcher asked two separate questions: **do higher petrol prices actually cause more theft**, and **what else in the economy is driving the level of theft over time?**

+3%

Rise in petrol thefts caused by a **1% rise in petrol prices**, cumulatively over the months that follow.

37

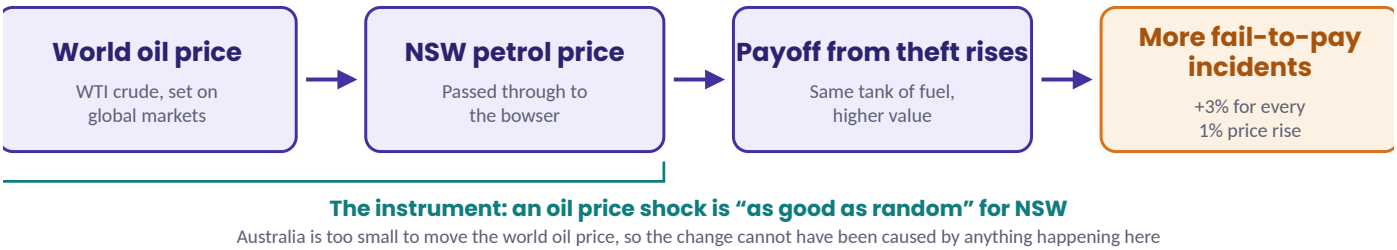
Extra thefts per month that implies for NSW, against a 2025 average of **1,118 incidents a month**.

6.5 vs 0.9

Medium-term elasticity of theft with respect to **living costs** versus **petrol prices**. Cost of living matters far more.

1 The economic logic being tested

Gary Becker's 1968 model of crime treats an offender as a rational decision-maker: an act is committed if the **expected benefit exceeds the expected cost**, where cost includes the probability and severity of punishment and the **opportunity cost** of the time involved. Petrol is an unusually clean test of that idea: the "payoff" from driving off without paying is simply the value of the tank, which rises automatically when the bowser price rises — with no change at all in the risk of being caught.



Why an economist can't just look at Figure 1

Earlier NSW research found a correlation of **0.9** between Sydney petrol prices and service station fraud (Moffatt & Fitzgerald, 2006). But a third factor — a recession, an inflation surge, population growth — would produce exactly that by pushing both series at once. **Causation** needs price variation nothing else in NSW could have caused.

2 Three methods, three different answers

The study runs three models. Each uses a **different source of price variation** and therefore answers a different question — reading them together is the whole point.

Two-way fixed effects
DAY-TO-DAY, LGA-BY-LGA

Do people react at the bowser the same day?

Compares daily price differences *between* 111 local government areas, holding each area and each date constant.

No significant effect. A 10c rise buys just 0.003 extra thefts a day.

Instrumental variables
MONTH-TO-MONTH, STATEWIDE

Do thefts respond to a genuine price shock?

Uses monthly swings in the **world oil price (WTI)** to isolate price changes that NSW conditions cannot have caused.

Yes. Thefts rise 1.7% immediately, then to about 3.3%.

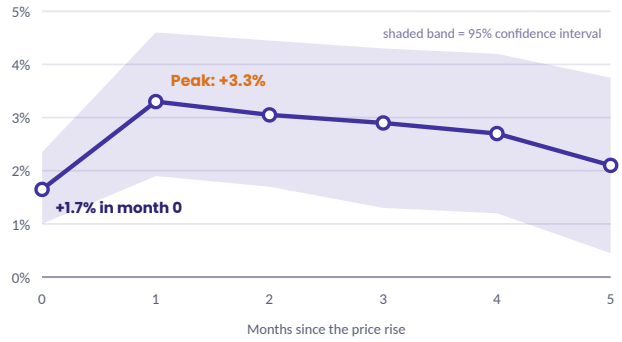
Error correction model
MEDIUM TERM, IN LEVELS

What does the level of theft track over eight years?

Models thefts alongside prices, living costs, wages and population.

Descriptive, not causal — and living costs and wages dominate.

Figure A · The causal effect of a 1% petrol price rise



Cumulative % change in monthly petrol thefts after a 1% price increase, instrumented with global oil prices. Adapted from Figure 6 and Appendix A of the bulletin.

Reading the causal result

The two short-run models look contradictory but aren't. The daily model finds nothing because **prices barely differ between LGAs** — retailers keep close to their competitors, so there is little variation left to explain — and because a theft plausibly requires some planning rather than a snap decision at the pump.

The monthly model, using a real price **shock**, does find an effect. Together they suggest the response is real but takes time to work through: it is a reaction to a sustained change in the cost of filling the tank, not to today's board price.

The statistics matter

The shaded band is the 95% confidence interval: clearly above zero early on, then widening. An estimate that is not statistically significant is **not evidence of no effect** — it may simply mean the data cannot detect one.

3 Over the medium term, prices are the small story

Figure B · What the level of theft moves with, 2016–25



Estimated % change in the level of petrol theft for a 1% higher level of each variable (error correction model, Table 3), all significant at the 1% level. **Historical associations, not causal effects.**

Holding everything else constant, a 1% higher level of petrol prices is associated with thefts about 1% higher. But a 1% higher **cost of living** is associated with thefts **6.5% higher**, and 1% higher **nominal wages** with thefts **16% lower**.

Read together, those two describe one thing: **real income**. Prices up and wages flat means purchasing power falls. That, more than the petrol price itself, is what theft has tracked — which resolves the 2022–24 puzzle, a period of strong living-cost growth but subdued petrol price growth.

4 Where this fits the HSC course

References are to the **Economics Stage 6 Syllabus (2009)**, the syllabus assessed in the 2026 and 2027 HSC examinations.

TOPIC 3 Economic Issues

- **Inflation — effects of inflation.** A measurable effect on decision-making: the elasticity on living costs (6.51) is **seven times** that on petrol prices. Rising general prices, not one price, is what the level of theft tracks.
- **Distribution of income and wealth.** The wages coefficient (−16.35), read alongside living costs, is a statement about **real** income — falling purchasing power, concentrated among low-income households.
- **Unemployment — social costs.** A useful complication: the unemployment rate was **small and not statistically significant**. Wages did the work instead, suggesting the pressure is on people who *are* employed but whose incomes are not keeping pace.
- **Economic growth and the business cycle.** Bushway et al. (2012), cited in the bulletin, find income-motivated crime rises during recessions.

TOPIC 1 The Global Economy

- **Globalisation — trade in commodities.** The study's causal strategy depends on Australia being a **price taker**: "Australia's supply of and demand for oil are small relative to the rest of the world." A small open economy, put to work as a research design.
- **Transmission of global shocks.** Figure 3 of the bulletin shows monthly changes in WTI crude and NSW unleaded 91 moving together — pass-through from a world commodity market to a household expense, and on to a domestic social outcome.

TOPIC 2 Australia's Place in the Global Economy

- **Exchange rates — effects of movements in the AUD.** The researcher measures oil in **US dollars** to avoid capturing exchange rate movements, noting that changes in the AUD can themselves affect petrol prices. Imported inflation in one sentence: a depreciation raises the AUD cost of oil, the bowser price, and (on these estimates) theft.

TOPIC 4 Economic Policies and Management

- **Monetary policy — objectives.** Price stability is usually defended in terms of certainty and real incomes. This study adds an empirical social cost to that case.
- **Fiscal policy and cost-of-living measures.** If living costs and real wages dominate, targeted income support does more for this outcome than fuel-specific measures such as the 2022 excise cut — a concrete way to argue policy effectiveness.
- **Limitations of policy — time lags.** The 39–46% monthly speed of adjustment is an **impact lag you can quote a number for**.
- **Microeconomic reform.** FuelCheck, the NSW Fair Trading price-disclosure scheme that supplied the data, is itself a market-transparency reform.

Skills: correlation is not causation

- **Find variation you can defend.** Not any price change — one caused by something outside the system.
- **State the assumptions.** The author names all three conditions the instrument must meet, and concedes the third weakens over time.
- **Say what you cannot know.** Omitted factors such as CCTV coverage are flagged, and re-running the model back to 2004 changes the relationships — a warning against generalising.

Looking ahead: the 2025 syllabus

The **Economics 11–12 Syllabus (2025)** starts in 2027, first examined in 2028. This material maps forward to all three Year 12 focus areas:

- **Economic issues in the Australian economy** — inflation, real incomes, distribution
- **Economic management of the Australian economy** — policy responses, lags, effectiveness
- **Australia and the global economy** — commodity price shocks and pass-through

5 Using the bulletin as exam stimulus

Treat pages 1–2 of this summary as the stimulus material.

Short answer

1. **Distinguish** between correlation and causation, using the relationship between petrol prices and petrol thefts as your example. — **3 marks**

2. **Explain** why the study uses changes in the world oil price, rather than changes in the NSW petrol price itself, to estimate the causal effect on theft. — **4 marks**

3. The study estimates a medium-term elasticity of 0.94 for petrol prices and 6.51 for living costs. **Analyse** what these two figures together suggest about the causes of the rise in petrol theft between 2022 and 2024. — **4 marks**

4. **Explain** how a depreciation of the Australian dollar could affect petrol prices in NSW, and **outline** one consequence for the level of petrol theft suggested by the study. — **5 marks**

5. The unemployment rate was found to be small and not statistically significant, while nominal wages had a large negative coefficient. **Discuss** what this implies about which households are under budget pressure. — **4 marks**

6. **Assess** the usefulness of this study for a government designing a cost-of-living policy response. — **4 marks**

Extended response — choose ONE

QUESTION 1 · 20 MARKS

Analyse the economic and social costs of inflation for the Australian economy, and **evaluate** the effectiveness of the policies available to government in addressing them. In your answer, refer to the stimulus.

QUESTION 2 · 20 MARKS

Explain how movements in world commodity prices are transmitted to the Australian economy, and **analyse** the impact of such movements on Australian households. In your answer, refer to the stimulus.

> **Marker's note**
>
> A Band 6 response does not simply quote "+3%". It uses the contrast — a small causal effect from prices, a much larger association with living costs and real wages — to build an argument about **which** economic pressure matters, and it acknowledges that the larger figures are descriptive rather than causal.

Key terms

Causal effect — the change in an outcome that a change in one variable actually produces, holding all else constant.

Instrumental variable — an outside factor (here, world oil prices) used to isolate variation in a cause that could not have been produced by the outcome.

Elasticity — the % change in one variable following a 1% change in another.

Real wages — nominal wages adjusted for price changes; a measure of purchasing power.

Error correction model — a model of how a variable returns to a medium-term path after a shock, and how quickly.

Statistical significance — confidence that an estimated effect is not simply the product of chance.

Price taker — an economy or firm too small to influence the world price of a good.

Opportunity cost — the value of the next best alternative forgone; in Becker's model, what an offender gives up by spending time on crime.

Source. Adams, N. (2026). *How do petrol thefts respond to changing prices?* (Crime and Justice Bulletin No. 280). Sydney: NSW Bureau of Crime Statistics and Research. Data: NSW Police COPS; NSW Fair Trading FuelCheck; ABS; U.S. Energy Information Administration. Theft counts are recorded "fail to pay" incidents at service stations, July 2004 – December 2025.